Notes on the Additional 16S Analysis Folder

A first pass through Data/16s_analysis

I took a look through the additional 16S folder. There is a lot of useful work here, and the folder is organized in a way that makes the different analysis pieces easy to find. These are just notes on places where the code may need a little cleanup before we rerun it in the current repository. I have not changed anything in this folder yet.

What is in the folder

- Separate sections for ANCOM, core microbiome, indicator analysis, top abundance, and figures.
- A saved phyloseq object plus the output tables and figures from the analyses.
- A README explaining the species abbreviations and the purpose of each section.
- All eight R scripts are syntactically valid and the saved phyloseq object loads successfully.

Things I noticed while reading the code

ANCOM sections

The ANCOM script looks like it was developed and run a section at a time. In the stored version, T1 is assigned and then replaced by TF before the model is run, after which results are assigned to both time points. The saved T1 and TF outputs are different, so the intended analyses were probably run interactively. We will want to reorganize this section before rerunning it so each species and time point is explicit.

Indicator analysis T0 subset

The T0 indicator section uses Treatment == 'T0'. Because T0 is stored in Time.Point, that line would currently select no samples. This looks like a simple column-name typo to correct when we rebuild the workflow.

Sample S3059

The source metadata identifies S3059 as a Pocillopora acuta sample from parent 301, High treatment, at T1. In the saved phyloseq object, its Type is recorded as 'S3059' and its time point as 'F1'. Since the scripts first keep Type == 'sample', S3059 is left out. We should correct that record and confirm whether the analysis should contain 196 rather than 195 experimental samples.

Paths and working directories

Some scripts use Hannah's original computer path, while others expect to be run from either the analysis root or a procedure folder. That is normal for code moved between computers, but we will need to switch everything to one project-relative path convention before rerunning it here.

Pavona figure input

The combined figure script uses read.csv('') as a placeholder because Pavona had no significant ANCOM taxa. That placeholder stops a full run. We can replace it with an empty table that has the expected columns.

A few analysis decisions to revisit together

- T1 and TF include repeated fragments from the same parents. The PERMANOVA and ANCOM sections currently do not account for Parent.ID, so we should decide how to carry the paired design into those analyses.
- The core-microbiome prevalence threshold differs by species and was chosen to return roughly five core taxa. That can work for a descriptive figure; we should just document that choice clearly or decide on a common threshold.
- Several analyses use random permutations or NMDS without an explicit seed. Adding seeds will make regenerated results easier to compare.
- T0 samples retain their original Ambient and High assignments even though the treatment had not started. Some sections combine them for T0 and others keep the labels, so we should make that convention consistent.

What looks good

- There are no duplicated sample names in the saved phyloseq object.
- Samples marked Type == 'sample' have complete parent, treatment, species, time-point, tank, and fragment metadata.
- The experimental samples in the saved object are rarefied to 3,000 reads.
- The existing output files are useful as a reference while we make the scripts reproducible.

Suggested next steps

- Preserve the current outputs as the historical version.
- Correct S3059 and establish one project-relative path setup.
- Rewrite the ANCOM section so every species and time-point contrast runs independently.
- Fix the T0 indicator subset and the empty Pavona figure input.
- Add seeds and decide how Parent.ID should be handled in repeated-measures analyses.
- Regenerate everything into a new output folder and compare it with the existing results before replacing anything.

None of this means the existing biological results are necessarily wrong. Most of what I found looks like the normal residue of interactive analysis, moving code between computers, and revising figures over time. The goal is simply to make the final version easy for both of us to rerun and understand.